

SOLITAX® sc Turbidity and Suspended Solids Sensors

Turbidity and Suspended Solids

Features and Benefits

Accurate, Color-independent Measurement

Hach's SOLITAX sc Turbidity and Suspended Solids Sensors are immune to shifts in color of activated sludge caused by variations in through-flow volume, climatic fluctuations, or other unexpected events. Unlike conventional systems, that need to be continually calibrated to compensate for color variation, the proven measurement technique used in the SOLITAX sc sensor is the only technology that provides measurement completely independent of color.

Self-cleaning Wiper Prevents Erroneous Values

The SOLITAX sc sensor's self-cleaning wiper prevents erroneous values and maintenance problems by preventing biological growth. Gas bubbles in the sample do not interfere with the measurement.

Excellent Correlation to Laboratory Analysis

Comparisons between laboratory analysis and continuous suspended solids measurement using SOLITAX sc sensors show an exceptional correlation to mixed liquor, settled and thickened sludge. Online measurement not only saves time on manual analysis but also provides critical real time TSS measurements that can be used to operate the plant more efficiently. (See *Correlation Chart* on page 2.)

Fully Serviceable Sensors

Conventional Turbidity and Suspended Solids sensors are potted and are discarded when they no longer function. SOLITAX sc sensors are serviceable so that the useful life of the sensor, in many cases, is doubled.

Easy One-point Calibration

Calibration of SOLITAX sc sensors is easy with a simple correction factor procedure to measure turbidity or suspended solids. Multipoint calibration is available for nonlinear samples.

Versatile Mounting Styles

SOLITAX sc sensors can easily be configured for many applications. Models designed for immersion into a tank are available in PVC plastic or stainless steel. An insertion style sensor for pipes is available in stainless steel. Installation kits for either style sensor are also available.



Hach SOLITAX sc sensors provide accurate, color-independent measurement of turbidity and suspended solids in drinking water, wastewater, and industrial process applications. A self-cleaning wiper prevents measurement error due to fouling. These sensors demonstrate excellent correlation with laboratory analysis so it's an exceptional tool for process control.

Multi-channel, Multi-parameter System

Any combination of two SOLITAX sc sensors can be installed using one Hach sc100 Controller. The same controller can also accommodate any combination of parameters, including suspended solids, turbidity, pH/ORP, dissolved oxygen, conductivity, and nitrate. All of Hach's model sc sensors are "plug and play" with no complicated wiring or set-up procedure necessary.

Communications—Multiple alarm/control schemes are available using three relays and two PID control outputs. Communications use analog 4-20 mA and digital MODBUS®/RS485, MODBUS®/RS232 protocols. (Other digital protocols are available. Contact your Hach representative for details.) Every sc100 controller is equipped with wireless communication through an infrared port.

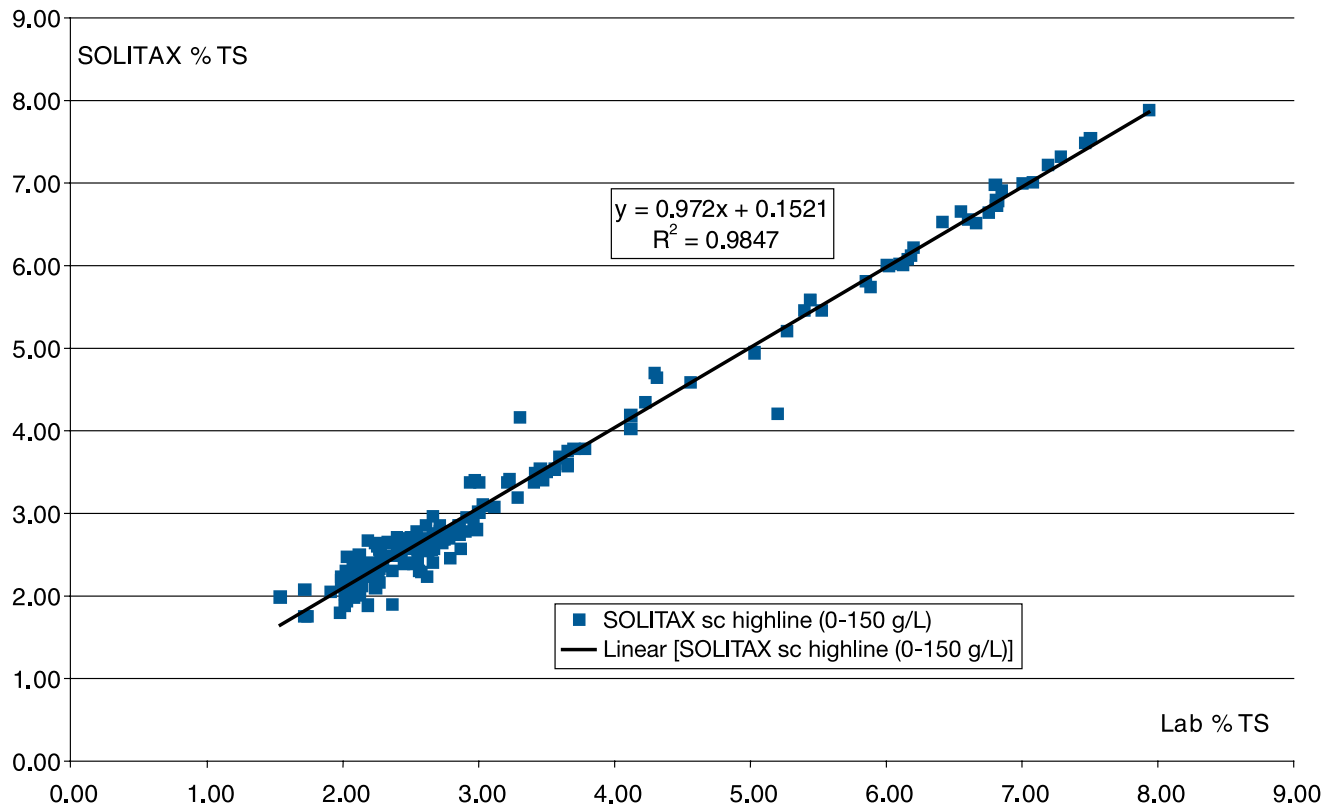
Data logger—A built-in data logger collects measurement data, calibration, verification points, and alarm history for up to 6 months. Data can be retrieved to a computer or PDA.

DW = drinking water WW = wastewater municipal PW = pure water / power
IW = industrial water E = environmental C = collections FB = food and beverage



Be Right™

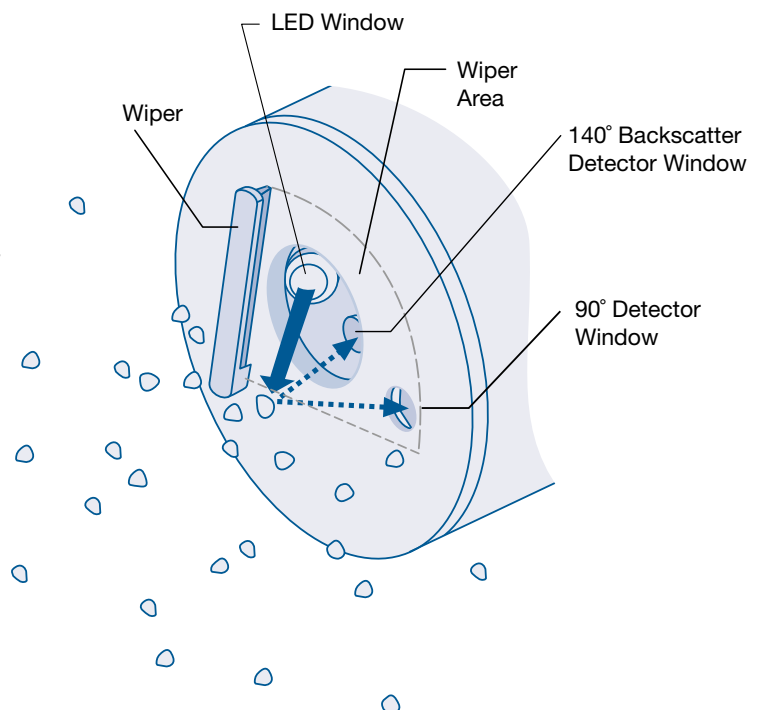
Correlation of SOLITAX sc *highline* sensor with lab analysis of sludge



Principal of Operation:

SOLITAX sc ts-line sensor with dual-beam optics and added back-scatter detector

- Dual infrared light beams. LED light source transmits light at 45° to sensor face.
- Nephelometric photoreceptors detect light at 90° to the transmitted light beam.
- Backscatter photoreceptor (included on all models except the SOLITAX sc t-line) detects light at 140° to the transmitted light beam to measure suspended solids in heavily loaded sample streams.
- Self-cleaning wiper, optional.
- T-line probes measure turbidity only.
- TS, HS, inline, and highline sensors measure either turbidity or suspended solids. The parameter to be measured is user selectable and can be changed at any time.



Specifications*

	For insertion in pipes		For immersion in open tanks		
	Model inline sc	Model highline sc	Model t-line sc	Model ts-line sc	Model hs-line sc
Parameter					
	Turbidity Suspended Solids	Turbidity Suspended Solids	Turbidity	Turbidity Suspended Solids	Turbidity Suspended Solids
Measuring Range					
<i>Turbidity Suspended Solids</i>	0.001 to 4000 NTU 0.001 mg/L to 50 g/L	0.001 to 4000 NTU 0.001 mg/L to 500 g/L	0.001 to 4000 NTU n/a	0.001 to 4000 NTU 0.001 mg/L to 50 g/L	0.001 to 4000 NTU 0.001 mg/L to 500 g/L
Units of Measure					
<i>Turbidity Suspended Solids</i>	User selectable—NTU, FNU, or TE/F User selectable—g/L, mg/L, ppm, or % solids				
Accuracy	Defined according to ISO/WD 13530.				
<i>Turbidity Suspended Solids</i>	Less than 1% of reading or ± 0.001 NTU, whichever is greater Less than 5% of reading (depends on homogeneity of municipal activated sludge)				
Repeatability	Defined according to ISO/WD 13530.				
<i>Turbidity Suspended Solids</i>	Less than 1% of reading Less than 3% of reading (depends on the homogeneity of municipal activated sludge)				
Detection Limit					
<i>Turbidity Suspended Solids</i>	0.001 NTU 0.001 mg/L				
Response Time	Initial response in 1 second				
Signal Average Time	User selectable ranging from 1 to 300 seconds				
Calibration					
<i>Turbidity Suspended Solids</i>	Formazin or StablCal® Standard Based on gravimetric TSS analysis with a correction factor procedure				
Certifications	CE certified to EN 61326-1, EN 61326/A1, EN 61326/A2, EN 61010-1				
Mounting					
	Through sidewall of a pipeline using a ball valve; minimum pipe size 100 mm (4 in.) in carbon or stainless steel		Fixed to the rim of the tank		
Flow Velocity	3 m/s (9.8 ft./s) maximum				
Operating Temperature	0 to 40°C (32 to 104°F)				
Sample Temperature	0 to 40°C (32 to 104°F)				
Sample Pressure	6 bar (87 psi) maximum				
Sensor Construction					
<i>Wiper Sensor body</i>	Silicon Stainless steel		Silicon Stainless steel or PVC		
Dimensions	(diameter x length)				
	Insertion sensor: 60 x 315 mm (2.4 x 12.4 in.); Ball valve from flange: 185 x 625 mm (7.3 x 24.6 in.)		Immersion sensor: 60 x 200 mm (2.4 x 7.9 in.)		
Weight					
	Insertion stainless steel: 2.4 kg (5.3 lb.)		Immersion, stainless steel: 1.38 kg (3.0 lb.) Immersion, PVC: 0.52 kg (1.2 lb.)		
Cable Length					
	10 m (33 ft.) standard. Optional extension cables available in 7.6 m (25 ft.), 15.2 m (50 ft.), 30.5 m (100 ft.). Maximum total length: 100 m (328 ft.).				

*Specifications subject to change without notice.

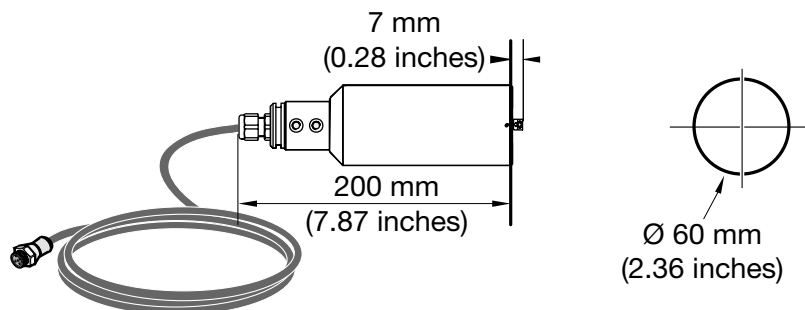
Engineering Specifications

1. The suspended solids and/or turbidity sensor shall consist of a digital sensor designed to connect to a universal controller.
2. The suspended solids and/or turbidity sensor shall use dual infrared light beam technique design utilizing an LED light source in the sensor to transmit an infrared/scattered dual beam into the sample stream at an angle of 45 degrees to the sensor face.
3. A photoreceptor located in the sensor face shall detect scattered light (turbidity) at 90 degrees to the transmitted beam.
4. A backscatter photoreceptor (included on all models except the SOLITAX sc t-line), positioned at 140 degrees to the transmitted beam, shall detect light scattered for suspended solids readings.
5. The sensor shall analyze the dual-beam data and provides a color independent measurement.
6. The sensor shall have a self-cleaning wiper to prevent erroneous values and maintenance problems caused by biological activity, scum build-up, and gas bubbles.
7. The self-cleaning wiper shall ensure that measurements are accurate, continuous, and completely color independent.
8. The sensor shall have an operating range of 0.001 mg/L to 50 g/L or 0.001 mg/L to 500 g/L and/or 0.001 to 4000 NTU suspended solids and turbidity depending on the model.
9. The sensor shall be capable of immersion in a tank or insertion into a pipe.
10. The materials of construction shall be PVC or polished stainless steel, depending on the sensor selection.
11. The sensor shall have an operating temperature of >0 to 40°C (32 to 104°F).
12. The initial response time shall be 1 second and user-adjustable up to 300 seconds.
13. The accuracy shall be less than 1% of reading or 0.01 NTU, whichever is greater for turbidity and less than 5% of reading for suspended solids.
14. The suspended solids and turbidity sensor shall be Hach Company Model t-line sc for turbidity measurement or ts-line sc, hs-line sc, inline sc, or highline sc for suspended solids and turbidity measurement.

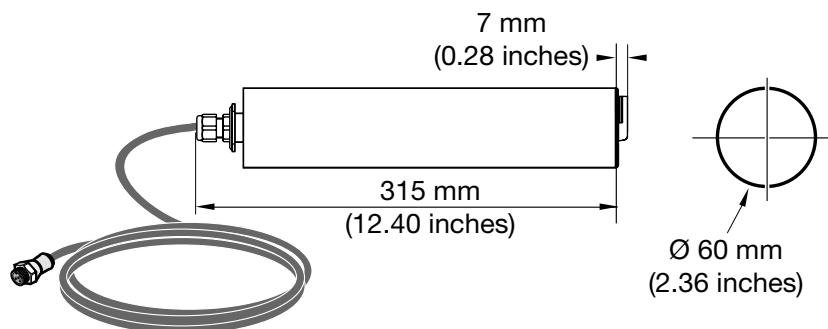
Dimensions

Hach SOLITAX sc sensors can be fixed to the rim of the tank for immersion applications or inserted directly through the sidewall of a pipeline for insertion applications. The optical unit of the sensor should be installed facing downstream in the direction of the flow to minimize the impact of debris against the measurement system. A variety of installation kits are available.

SOLITAX sc models t-line, ts-line, and hs-line sensors for immersion in open tanks

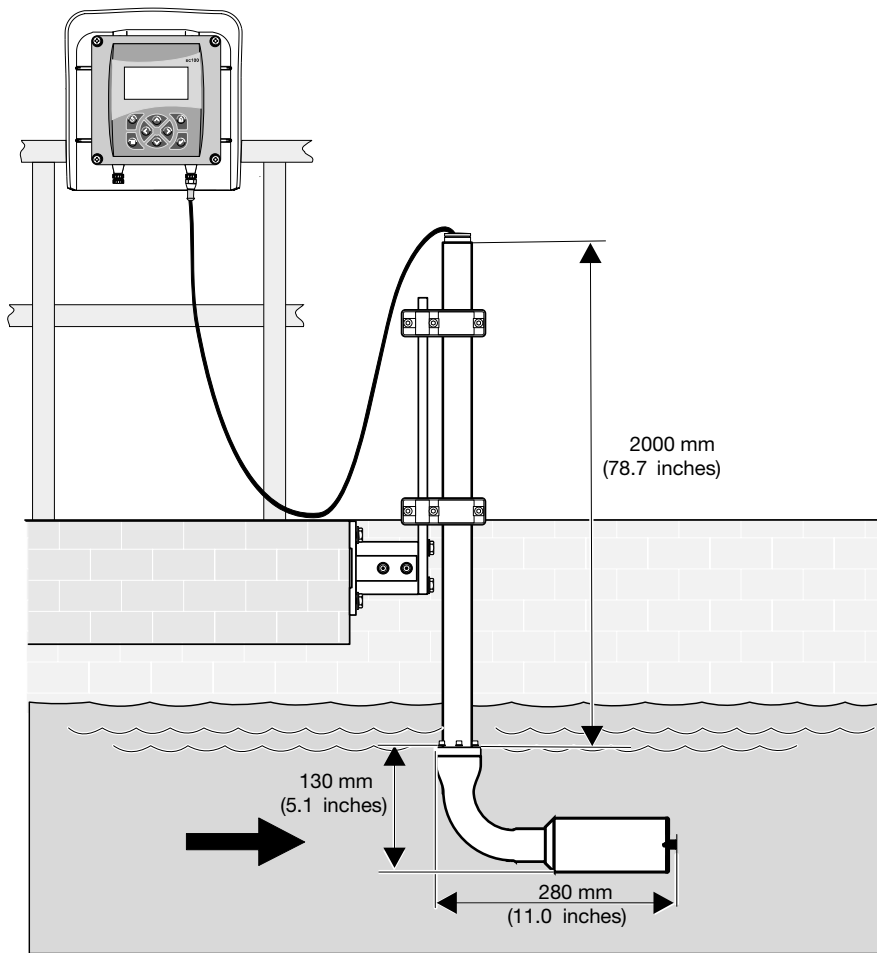


SOLITAX sc models inline and highline sensors for insertion in pipes

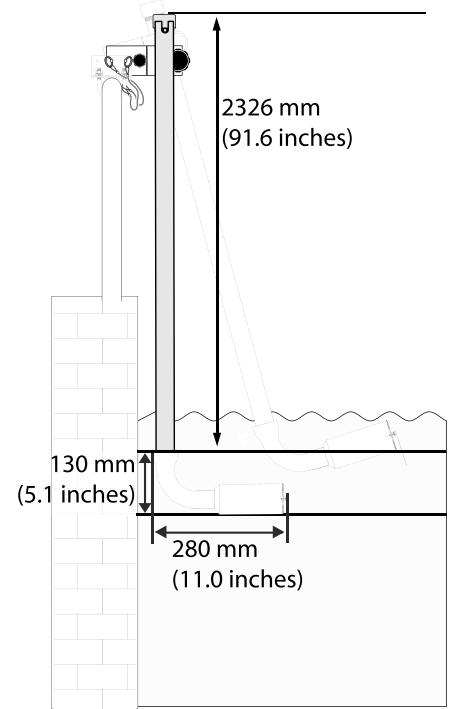


Dimensions continued

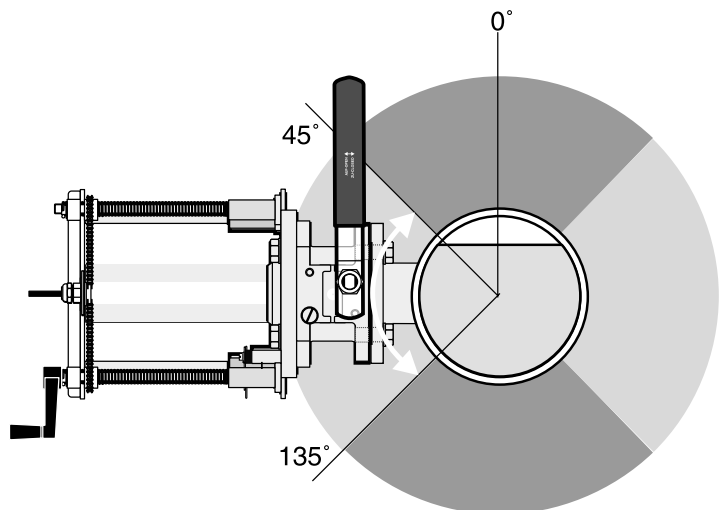
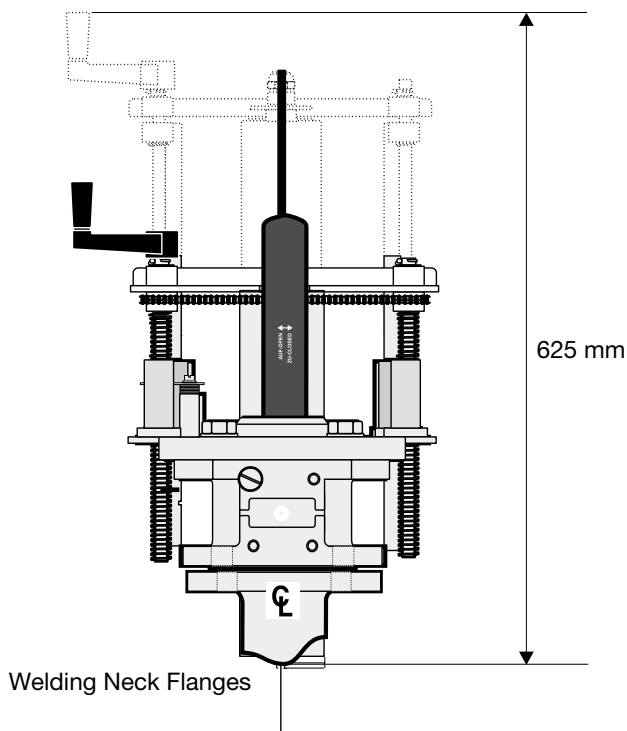
Installation for mounting SOLITAX sc models t-line, ts-line, and hs-line sensors for immersion in open tanks. (Fixed point Installation Kit, P/N 5734400)



Handrail Mounting Kit includes 1.5-inch diameter by 7.5-ft long CPVC pipe and swivel/pivot/ pipe clamp assembly (P/N MH236B00Z)



Fixture with ball valve for mounting SOLITAX sc models inline and highline sensors in pipes



Ordering Information

Sensor Selection

		For insertion in pipes		For immersion in open tanks		
	Range	inline	highline	t-line	ts-line	hs-line
Wastewater						
Final effluent	0 to 0.1 g/L or 0 to 1000 NTU	●		●	●	
Influent suspended solids	0 to 1.5 g/L	●			●	●
Filtrate/centrate	0 to 2 g/L	●			●	●
Mixed liquor suspended solids	0 to 5 g/L	●			●	
Return activated sludge	0 to 25 g/L	●			●	
Waste activated sludge	0 to 2% or 0 to 20 g/L	●			●	
Primary sludge (settled sludge)	0.5 to 10% or 5 to 100 g/L		●			●
Anaerobic digested sludge	1.5 to 7% or 15 to 70 g/L		●			●
Gravity thickened sludge	2 to 10% or 20 to 100 g/L		●			●
Mechanically thickened sludge	3 to 8% or 30 to 80 g/L		●			●
Drinking Water						
Source water monitoring	0 to 500 NTU	●		●	●	
Influent pressurized raw water	0 to 500 NTU	●			●	●
Clarifier efficiency	0 to 20 NTU	●		●	●	
Filter backwash	0 to 20 NTU	●		●	●	
De-watering	0.5 to 10% or 5 to 100 g/L	●	●			●
Industrial Water						
Make-up water—Power, Pulp & Paper, Petrochemical, Food & Beverage		●		●	●	
Boiler and cooling water—Petrochemical, Pulp & Paper				●	●	
Lime addition—Petrochemical, Mining, Pulp & Paper			●			●
Wastewater (activated sludge process)—Power, Pulp & Paper, Petrochemical, Food & Beverage		●	●	●	●	●
White liquid applications—Pulp & Paper			●			●
Stormwater—Pulp & Paper, Petrochemical		●		●	●	

Ordering Information *continued*

SOLITAX sc Turbidity and Suspended Solids Analyzers

(common controller and sensor configurations)

Immersion in Open Tanks Applications

- 6940000** Turbidity Analyzer includes sc100 controller and PVC t-line sc sensor (0.001 to 4000 NTU) with wiper
- 6940100** Turbidity and Suspended Solids Analyzer includes sc100 controller and stainless steel ts-line sc sensor (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L) with wiper
- 6940200** Turbidity and High Range Suspended Solids Analyzer includes sc100 controller and stainless steel hs-line sc sensor (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L) with wiper

Insertion in Pipes Applications

- 6940300** Turbidity and Suspended Solids Analyzer includes sc100 controller, stainless steel inline sc sensor (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L) with wiper, and insertion mounting kit
- 6940400** Turbidity and High Range Suspended Solids Analyzer includes sc100 controller, stainless steel highline sc sensor (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L) with wiper, and insertion mounting kit

Both 6940300 and 6940400 include the insertion mounting kit which includes a 4 inch pre-coped Carbon Steel Flange. Non-coped flanges are available. See below.

NOTE

1. Power cords must be ordered separately.
2. Fixed point installation kit or handrail mount kit must be ordered separately for all immersion analyzers.

Individual SOLITAX sc Sensors

Immersion Sensors

- LXV423.99.10000** Turbidity, t-line sc, PVC with wiper (0.001 to 4000 NTU)
- LXV423.99.12000** Turbidity, t-line sc, PVC without wiper (0.001 to 4000 NTU)
- LXV423.99.10100** Turbidity and Suspended Solids, ts-line sc, PVC with wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV423.99.12100** Turbidity and Suspended Solids, ts-line sc, PVC without wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV423.99.00100** Turbidity and Suspended Solids, ts-line sc, stainless steel with wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV423.99.02100** Turbidity and Suspended Solids, ts-line sc, stainless steel without wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV423.99.10200** Turbidity and Suspended Solids, hs-line sc, PVC with wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)
- LXV423.99.12200** Turbidity and Suspended Solids, hs-line sc, PVC without wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)
- LXV423.99.00200** Turbidity and Suspended Solids, hs-line sc, stainless steel with wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)
- LXV423.99.02200** Turbidity and Suspended Solids, hs-line sc, stainless steel without wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)

Insertion Sensors

- LXV424.99.00100** Turbidity and Suspended Solids, inline sc, stainless steel with wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV424.99.02100** Turbidity and Suspended Solids, inline sc, stainless steel without wiper (0.001 to 4000 NTU, 0.001 mg/L to 50 g/L)
- LXV424.99.00200** Turbidity and Suspended Solids, highline sc, stainless steel with wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)
- LXV424.99.02200** Turbidity and Suspended Solids, highline sc, stainless steel without wiper (0.001 to 4000 NTU, 0.001 mg/L to 500 g/L)

Accessories

- 5733000** Calibration Kit, includes calibration cylinder, two 500-mL 800 NTU StablCal®, and a sensor bracket
- 2660549** 800 NTU StablCal®, 500-mL (two required per calibration)
- LZX050** Wiper Blades, replacement only, pkg. of 5
- LZX961.54** Sun Shield for sc100 controller

Cable Accessories

- 5867000** Junction Box (for extension cables*)
- 5796000** 7.6 m (25 ft.) Extension Cable
- 5796100** 15.2 m (50 ft.) Extension Cable
- 5796200** 30.5 m (100 ft.) Extension Cable
- *Maximum total length 100 m (328 ft.)*

Installation Accessories

- 5734400** Fixed Point Installation Kit for t-line, ts-line, and hs-line immersion sensors, includes stand and sun shield for controller and pipe for sensor cable
- 5738400** Insertion Mounting Kit for inline and highline insertion sensors (ball valve and extraction system). Kit includes a 4 inch pre-coped Carbon Steel Flange. Non-coped flanges are available. See below.
- AHA033NPT** Sensor Adapter, straight 1-1/2 FNPT
- AHA034NPT** Sensor Adapter, elbow 1-1/2 FNPT, 90°
- MH236B00Z** Handrail Mounting Kit (for sensor to be used with either adapter above) includes 1.5-inch diameter by 7.5-ft long CPVC pipe and swivel/pivot/ pipe clamp assembly
- LZX660** Non-coped stainless steel welding flange for insertion kit
- LZX661** Non-coped carbon steel welding flange for insertion kit

To complete your turbidity and suspended solids measurement system, choose from these Hach controllers...

Model sc100 Controller

(see Lit. #2463)

The Model sc100 Controller receives data from one or two sensors. Its "plug and play", mix-and-match operation lets it fit into any facility or workflow. Digital communication with any Hach digital sensor or probe is simple and reliable.



Model sc1000 Controller

(see Lit. #2403)

Get the same great features as the sc100 Controller above—"plug and play", all digital operation and communication—but with the Hach sc1000 Controller, up to eight Hach sensors can be used with one controller in any combination. The sc1000 Controller is also expandable and upgradeable to easily adapt to your needs.



At Hach, it's about learning from our customers and providing the right answers. It's more than ensuring the quality of water—it's about ensuring the quality of life. When it comes to the things that touch our lives...

Keep it pure.

Make it simple.

Be right.

For current price information, technical support, and ordering assistance, contact the Hach office or distributor serving your area.

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